C 81806 (Pages : 3) Name.....

Reg. No.....

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, APRIL 2020

Mathematics

MAT 2C 02—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type)

Answer **all twelve** *questions.*
Each question carries 1 *mark.*

1. Find a formula for nth term of the sequence 1, –1, 1, –1,.....
2. Define a non-decreasing sequence.
3. Write $\tanh x$ in terms of exponential function.
4. State Sandwich theorem for sequences.
5. Find the domain of the function $w = \sqrt{y - x^2}$.
6. Define contour line of a function $f(x,y)$.
7. Find $\lim_{(x,y)\to(0,1)} \frac{2 - xy + 3}{x^2 y + 5xy - y^3}$.
8. Define absolute convergence of a series.
9. $\frac{d}{dx}\sinh 2x =$ _____.
10. Define level curve of a function.
11. Find $\lim_{n\to\infty} n^{1/n}$.
12. $\int_0^1 \sinh x dx =$ _____.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 *marks.*

13. Is the area under the curve $y = 1/\sqrt{x}$ from $x = 0$ to $x = 1$ finite ? If so, what is it ?
14. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1 + x^2}$.

Turn over

15. Show that $\lim_{n\to\infty}\frac{1}{n}=0$.

16. Determine whether the sequence $a_n=\frac{2^n3^n}{n!}$ is non-decreasing and bounded from above.

17. Show that $f(x,y)=\begin{cases}\frac{2xy}{x^2+y^2}, & (x,y)\neq(0,0);\\ 0, & (x,y)=(0,0);\end{cases}$ is continuous at every point except the origin.

18. Find the values of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the point (4,–5) if $f(x,y)=x^2+3xy+y-1$.

19. Find the volume of the solid generated by revolving the region between y-axis and the curve $x=2/y, 1\le y\le 4$ about the y-axis.

20. Find $\int_0^{ln2}4e^x\sin x dx$.

21. Graph the sets of points whose polar co-ordinates satisfies the conditions $1\le r\le 2$ and $0\le\theta\le\pi/2$.

22. Replace the polar equation $r=\frac{4}{2\cos\theta-\sin\theta}$ equivalent cartesian equation.

23. Find the spherical co-ordinates equation for the cone $z=\sqrt{x^2+y^2}$.

24. Find $\frac{dw}{dt}$ if $w=xy+z, x=\cos t, y=\sin t, z=t$. What is the derivative's value at $t=0$?

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Investigate the convergence of $\int_0^3\frac{dx}{(x-1)^{2/3}}$.

26. Show that $(-1)^n\frac{n-1}{n}$ diverges.

27. Find the area of the region in the plane enclosed by the cardioid $r=2(1+\cos\theta)$.

28. Show that the p-series $\sum_{n=1}^{\infty}\frac{1}{n^p}$ converges if $p>1$ and diverges if $p\le 1$.

29. Find the linearization of $f(x,y,z) = x^2 - xy + 3\sin z$ at the point (x_0, y_0, z_0).

30. Find $\frac{dw}{dt}$ if $w = xy + z, x = \cos t, y = \sin t, z = t$. What is the derivative's value at $t = 0$?

31. Find the area of the region that lies inside the circle $r = 1$ and outside the cardioid $r = 1 - \cos\theta$.

32. Show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 z}{\partial z^2} = 0$ if $f(x,y,z) = \left(x^2 + y^2 + z^2\right)^{-1/2}$.

33. Find the radius and interval of convergence of the series $\sum_{n=0}^{\infty} (-1)^n (4x+1)^n$.

(6 × 5 = 30 marks)

Part D (Essay Type)

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Evaluate $\int_2^{\infty} \frac{x+3}{(x-1)\left(x^2+1\right)} dx$.

35. Find the length of the curve $y = (x/2)^{2/3}$ from $x = 0$ to $x = 2$.

36. Find the lateral surface area of the cone generated by revolving the line segment $y = x/2, 0 \le x \le 4$ about the x-axis.

(2 × 10 = 20 marks)